Roboshot Procedure 2016_031	Oct 18th 2016
Procedure for Necessary Data collection for failure analysis [Investigation for the failed parts on ROBOSHOT]	Applicable for All Models (A07B-****-B***#****)

Necessary Data of Investigation for the failed parts on ROBOSHOT

Equipment

Model	**FANUC ROBOSHOT (all models)**
Specification	**Refer Below details**
Spec. of mechanical unit assembly	**A07B-****-B***#******

In the case of investigation needed for the failed parts send the faulty parts with the specific data as below.

1) Parts requiring data acquisition

On failure of the parts (unit) shown in Table 2, in the case of investigation by FANUC JAPAN needed, forward the failed parts with the acquired data shown in Table 4.

Table 2 -Parts requiring data acquisition

X axis	Motor, Ball Screw, Bearing, Pulley, Belt
C axis	Motor, Bearing, Pulley, Belt
Y axis	Motor, Ball Screw, Bearing, Pulley, Belt
A	Motor, Ball Screw, Bearing, Pulley, Belt
Z	Motor, Ball Screw, Bearing
B axis	Motor
Clamp unit	Toggle pin, Toggle bush, Strain rod

2) Tools and Measuring instruments needed

Prepare general working tools and measuring instruments corresponding to the failed axis as shown in below Table 3

Table 3 : Tools and Measuring instruments

			Failed Axis							
No	**Name**	**Specification etc.,**	**X**	**C**	**Y**	**A**	**Z**	**B**	**Clamp Unit**	**Q'ty**
1	Tools	General working tools	○	○	○	○	○	○		1
2	Belt tension meter		○	○	○	○				1
3	Thickness gauge	thickness = 0.01mm	○	○	○	○				1
4	Depth gauge	1 DIV.= 0.01mm				○				1
5	Vernier calliper	1 DIV.= 0.05mm or 0.01mm					○			1
6	Clamping force sensor	Ring type						○	○	-
7	Inside micrometre	1 DIV. = 0.01mm						○	○	1
8	Leveller	1 DIV.=0.1 mm/M or 0.05mm/M						○		
9	Memory card	Compact flash card	○	○	○	○	○	○	○	1
10	Toggle pin extractor (Jig)	Confirm that your working jig is good for the maintenance of the failed machine.							○	1
11	Toggle bush remover (Jig)								○	1
12	Set-up for toggle bush (Jig)								○	1
13	Strain rod extractor (Jig)								○	1
14	Dry ice for toggle bush expansion fit	About 10 kg prepared							○	1
15	Dummy mold	High precision in flatness and parallelism, such as a test block						○	○	1

3) Contents of work

Acquire the data corresponding to the failed parts (unit), referring to Table 4. And forward the data with the failed parts (unit) for investigation to FANUC JAPAN

1). Item No.2,6, 8, 13, 14 in Table 4 Save Data and SRAM data (on and after S-2000iB) or SRAM data (on and before S-2000iA) to the memory card. And necessary data corresponding to item No.2 - 6, 8, 13, 14 in Table 4 will be acquired.

2). Item No.7 in Table 4 Measure the belt tension by the use of belt tension meter before disassembling.

3). Item No.10 in Table 4 In case of the flange of drive pulley detached down, send it with the failed drive pulley.

4). Item No.11 in Table 4 In case of a gap found between the key on the shaft and the keyway on the pulley, measure the gap by the use of thickness gauge before disassembling

5). Item No.17 in Table 4 After returning to zero-point of A-axis on the screen, measure the distance between the mold face of MV. Platen and the tip of ejector pin by the use of depth gauge (or Vernier calliper).

6). Item No.19 in Table 4 If nozzle touching can be made, measure the compression stroke (actuating stroke of proximity switch). If not, confirm that the witness mark indicating the position of the proximity switch is put

7). Item No.20 in Table 4 If removing the nozzle touch springs can be made, measure the natural lengths of nozzle touch spring by the use of Vernier calliper

8). Item No.26 in Table 4 If locking-up by the mold loaded on failure can be made, measure the clamping force and calculate the balance of clamping force. If not, confirm that the witness marks indicating the positions of the die height gear are put on the original positions or not.

9). Item No.27 in Table 4 If locking-up by the mold loaded on failure can be made, measure parallelism of two platens for the states of locking-up and 30mm-opening of mold. If not, measure parallelism only for the state of 30mm-opening of mold.

10). Item No.28 in Table 4 Measure the base levels on the same positions as the initial installation in the customer's site, by the use of leveller listed in Table 3.

11). Items No.30, No.31 in Table 4 Check all bolts affecting the failed axis for looseness. Check all bolts check-painted for deviation from initial state.

12). Items No.32, No.33 in Table 4
1) Strain-rod, tie-rod Take photographs of the sliding face.
2) Strain-rod bush, tie-rod bush
Measure the inner diameter of bush. Or measure the clearance to the strain-rod or the tie-rod by

12). The others in Table 4
For the other items, acquire the data by observation, photo taking and discussing with the customer.

4) Table 4 (part 1) Data acquisition items (X-, C-, Y- axes)

		X axis			C axis			Y axis			ON and AFTER S-2000iB		ON and BEFORE S-2000iA
		MOTOR	BS BEARING	BELT PULLEY	MOTOR	BEARING	BELT PULLEY	MOTOR	BS BEARING	BELT PULLEY	Fdata	SRAM	SRAM
1	MACHINE MODEL, SERIAL NUMBER, MANUFACTURING DATE, TOTAL SHOT NUMBER, CYCLE TIME	○	○	○	○	○	○	○	○	○			
2	SETTING CONDITION ON SCREEN	○	○	○	○	○	○	○	○	○		■	■
3	MAINTENANCE HISTORY	○	○	○	○	○	○	○	○	○	■		■
4	OPERATION HISTORY	○	○	○	○	○	○	○	○	○	■		■
5	ALARM HISTORY	○	○	○	○	○	○	○	○	○	■		■
6	GRAPH DATA FOR ACTUAL PRESSURE, ACTUAL VELOCITY AND LOAD CURRENT ETC.	○	○	○	○		○					■	■
7	BELT TESION ON FAILURE	○	○		○	○		○	○				
8	PLUGGED NOZZLE AND FREQUENCY	○	○								■		■
9	LUBRICANT, MAINLY APPLIED (CARTRIDGE GREASE)		○			○			○				
10	FAILURE ON THE FLANGE OF DRIVE PULLEY			○			○			○			
11	WEAR AND RATTLE ON KEYS OR SPLINES			○			○			○			
12	APPIED MATERIAL				○	○	○						
13	SETTING TEMPERATURE				○	○	○					■	■
14	SETTING CLAMPING FORCE							○	○	○		■	■
15	MOLD WEIGHT, MOLD STRUCTURE, BUILT-IN SPRINGS							○	○	○			
16	PARALLELISM OF 2 BALL-SCREWS (CHECK THE WITNESS MARKS ON BELT-PULLEYS)												
17	POSITIONING ACCURACY OF A-AXIS ZERO POINT												
18	BELT JUMPING (CHECK THE WITNESS MARKS ON BELT-PULLEYS)		○						○				
19	COMPRESSION STROKE OF NT SPRING (ACTUATING STROKE OF PROXIMITY SWITCH)												
20	NATURAL LENGTHS (MEASURED VALUES) OF TWO NT SPRINGS AND DIFFERENCE												
21	FAILURE OF NUTS FOR MATCHED BEARING ON BS SHAFT-END												
22	TAP CONDITION ON ST. PLATEN, RETAINING BS SHAFT-END												
23	FREQUENCY OF AUTOMATIC/MANUAL DIEHEIGHT ADJUSTMENT												
24	LUBRICATION CONDITION OF STRAIN-ROD TRAPEZOIDAL THREADS AND REAR NUT-SEATING FACE												
25	LUBRICATION CONDITION OF SLIDING PART UNDER REAR-PLATEN												
26	CLAMPING FORCE BALANCE												
27	PARALLELISM (LOCK-UP, 30MM-OPEN)												
28	BASE LEVELLING												
29	NOISE OR AN UNUSUAL SOUND						○			○			
30	CHECK BOLTS FOR LOOSENESS	○	○	○	○	○	○	○	○	○			
31	CHECK BEARING NUTS FOR LOOSENESS		○			○			○				
32	WEAR OF STRAIN-ROD, STRAIN-ROD BUSH												
33	WEAR OF TIE-ROD, TIE-ROD BUSH	○	○	○	○	○	○						

5) Table 4 (part 2) Data acquisition items (A-, Z-, B- axes, Clamp unit)

		A axis			Z axis		B axis	CLAMP UNIT		ON and AFTER S-2000iB		ON and BEFORE S-2000iA
		MOTOR	BS BEARING	BELT PULLEY	MOTOR	BS BEARING	MOTOR	TOGGLE PIN TOGGLE BUSH	STRAIN ROD	Fdata	SRAM	SRAM
1	MACHINE MODEL, SERIAL NUMBER, MANUFACTURING DATE, TOTAL SHOT NUMBER, CYCLE TIME	○	○	○	○	○	○	○	○			
2	SETTING CONDITION ON SCREEN	○	○	○	○	○		○	○		■	■
3	MAINTENANCE HISTORY	○	○	○	○	○	○	○	○	■		■
4	OPERATION HISTORY	○	○	○	○	○	○	○	○	■		■
5	ALARM HISTORY	○	○	○	○	○	○	○	○	■		■
6	GRAPH DATA FOR ACTUAL PRESSURE, ACTUAL VELOCITY AND LOAD CURRENT ETC.										■	■
7	BELT TESION ON FAILURE	○	○									
8	PLUGGED NOZZLE AND FREQUENCY									■		■
9	LUBRICANT, MAINLY APPLIED (CARTRIDGE GREASE)		○			○	○	○	○			
10	FAILURE ON THE FLANGE OF DRIVE PULLEY			○								
11	WEAR AND RATTLE ON KEYS OR SPLINES			○								
12	APPIED MATERIAL											
13	SETTING TEMPERATURE										■	■
14	SETTING CLAMPING FORCE							○	○		■	■
15	MOLD WEIGHT, MOLD STRUCTURE, BUILT-IN SPRINGS	○	○	○	○	○	○	○	○			
16	PARALLELISM OF 2 BALL-SCREWS (CHECK THE WITNESS MARKS ON BELT-PULLEYS)		○									
17	POSITIONING ACCURACY OF A-AXIS ZERO POINT	○	○	○								
18	BELT JUMPING (CHECK THE WITNESS MARKS ON BELT-PULLEYS)		○									
19	COMPRESSION STROKE OF NT SPRING (ACTUATING STROKE OF PROXIMITY SWITCH)				○	○						
20	NATURAL LENGTHS (MEASURED VALUES) OF TWO NT SPRINGS AND DIFFERENCE					○						
21	FAILURE OF NUTS FOR MATCHED BEARING ON BS SHAFT-END					○						
22	TAP CONDITION ON ST. PLATEN, RETAINING BS SHAFT-END					○						
23	FREQUENCY OF AUTOMATIC/MANUAL DIEHEIGHT ADJUSTMENT						○		○			
24	LUBRICATION CONDITION OF STRAIN-ROD TRAPEZOIDAL THREADS AND REAR NUT-SEATING FACE						○					
25	LUBRICATION CONDITION OF SLIDING PART UNDER REAR-PLATEN						○					
26	CLAMPING FORCE BALANCE						○	○				
27	PARALLELISM (LOCK-UP, 30MM-OPEN)						○	○	○			
28	BASE LEVELLING						○	○	○			
29	NOISE OR AN UNUSUAL SOUND											
30	CHECK BOLTS FOR LOOSENESS	○	○	○	○	○	○	○	○			
31	CHECK BEARING NUTS FOR LOOSENESS		○			○						
32	WEAR OF STRAIN-ROD, STRAIN-ROD BUSH						○	○	○			
33	WEAR OF TIE-ROD, TIE-ROD BUSH											

6) SUPPLEMENTARIES

1). On failure of the motor, the ball-screw, the bearing, the belt or the pulley, in the case of requests of investigation for the failed part (unit) to FANUC JAPAN, examine the failed machine for the items indicated "○ " in Table 4.

2). The data of items indicated "■" are included in Data or SRAM data

Forward the failed parts (unit), the examination results for the items indicated "○" in Table 4, Data and SRAM data altogether.